

Bhutan early warning system for climate change

Building ClimateSense: Leveraging AI to Combat Climate Change in Bhutan

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8/16/2025

Project overview

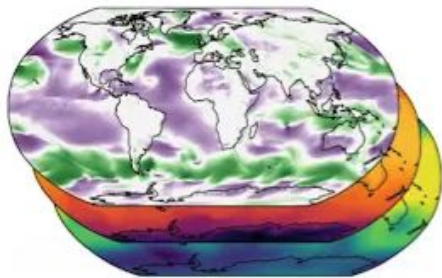
- Two active lines of work:
 - Weather forecasting: currently in deployment
 - ECMWF forecast for the next 15 days
 - GraphCastGFS: forecast for the next 16 days
 - Flood forecasting: currently in machine learning modeling phase, next step: deployment

GraphCast Global Forecast System (GraphCastGFS)

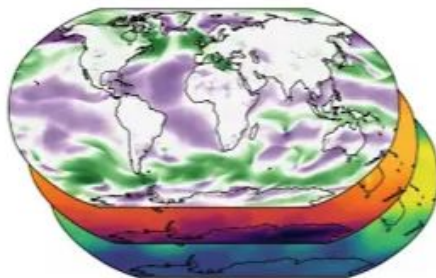
- Experimental medium-range global forecast system by National Centers for Environmental Prediction (NCEP).
 - Horizontal resolution: 0.25° latitude-longitude grid (~28 km).
 - Runs 4 times/day (00Z, 06Z, 12Z, 18Z) with forecasts every 6 hours.
 - Data format: GRIB2; based on ECMWF ERA5 reanalysis.
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- Update (v2.1, July 10, 2024): forecast range extended from 10 to 16 days.

GraphCast Global Forecast System (GraphCastGFS)

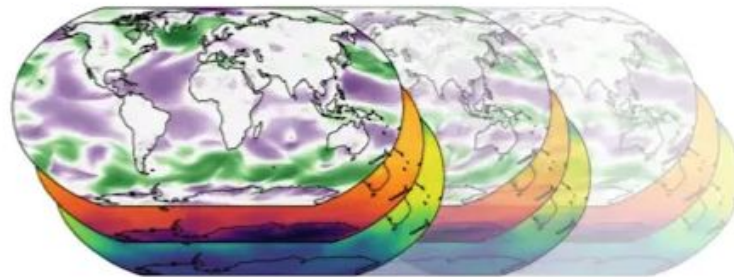
a) Input weather state



b) Predict the next state



c) Roll out a forecast



- Built on Google DeepMind's GraphCast ML model (GNN with encoder–processor–decoder architecture, ~37M parameters).
- Takes NCEP GDAS analysis data as input (current & 6-hr previous states).

ECMWF

Total variables found: 46

- 10 metre U wind component
- 10 metre V wind component
- 100 metre U wind component
- 100 metre V wind component
- 2 metre dewpoint temperature
- 2 metre temperature
- Divergence
- Eastward surface sea water velocity
- Geopotential
- Geopotential height
- Land-sea mask
- Maximum 10 metre wind gust since previous post-processing
- Maximum temperature at 2 metres in the last 3 hours
- Mean sea level pressure
- Minimum temperature at 2 metres in the last 3 hours
- Most-unstable CAPE
- Northward surface sea water velocity
- Precipitation type
- Relative humidity
- Runoff
- Sea ice thickness
- Sea surface height
- Skin temperature
- Slope of sub-gridscale orography
- Snow albedo
- Soil temperature
- Specific humidity
- Standard deviation of sub-gridscale orography
- Surface long-wave (thermal) radiation downwards
- Surface net long-wave (thermal) radiation
- Surface net short-wave (solar) radiation
- Surface pressure
- Surface short-wave (solar) radiation downwards
- Temperature
- Time-integrated eastward turbulent surface stress
- Time-integrated northward turbulent surface stress
- Top net long-wave (thermal) radiation
- Total column vertically-integrated water vapour
- Total column water
- Total precipitation
- Total precipitation rate
- U component of wind
- V component of wind
- Vertical velocity
- Volumetric soil moisture
- Vorticity (relative)

GraphCastGFS

Summary of contents:

- **Dataset 0:** `u10`, `v10` (10-meter wind components)
- **Dataset 1:** `t2m` (2-meter temperature)
- **Dataset 2:** Multiple pressure-level variables: `t`, `u`, `v`, `q`, `w`, `gh`
- **Dataset 3:** `prmsl` (mean sea level pressure)
- **Dataset 4:** `tp` (total precipitation)

All variables (except Dataset 2) share the same 2D spatial grid: `latitude` (721) × `longitude` (1440).

Dataset 2 includes pressure levels (`isobaricInhPa` : 13 levels) in addition to spatial dimensions.

Weather forecast: questions for discussion

- Have there been initiatives in Bhutan using ECMWF or AI-based ML models (e.g., GraphCast) for weather forecasting?
- Which weather variables are most critical to include?
- We are exploring interpolation methods to derive finer spatial detail from gridded data—any recommended approaches?

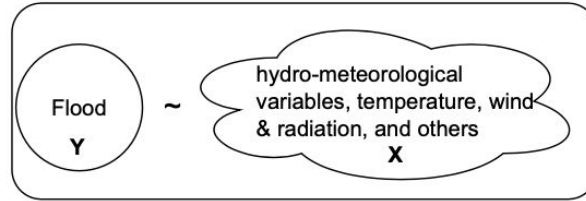
Project overview

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 - Weather forecasting: currently in deployment
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 - **Flood forecasting: currently in machine learning modeling phase, next step: deployment**
 - **Overview of ML workflow**
 - **Flood forecasting: scope**
 - **Data assembled so far**
 - **Spatial alignment challenge**

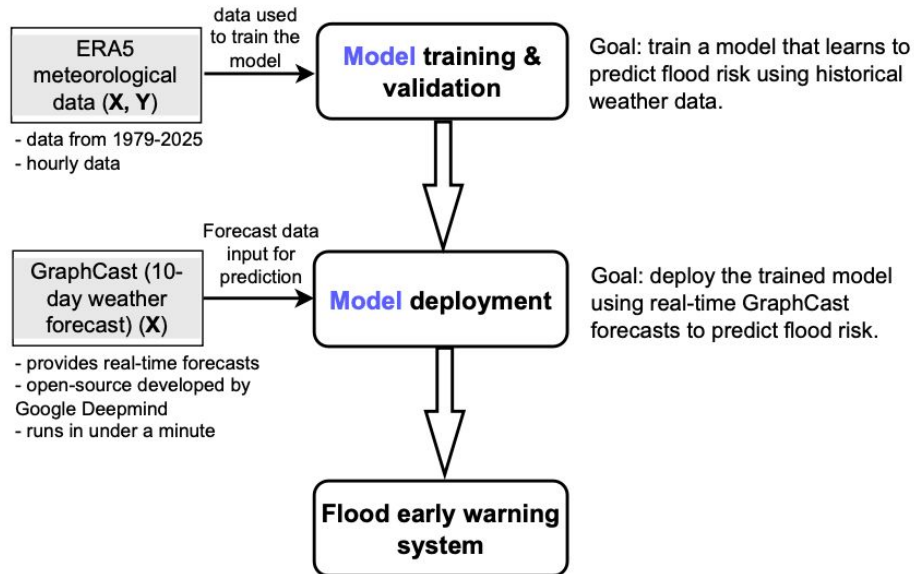
Bhutan Flood Risk Prediction System using machine learning

We train a flood risk model using decades of ERA5 data, and use GraphCast forecasts to deploy it in real time.

Machine learning **model**:
learns the relationship
between **X** and **Y**



*X = static features
(geospatial) + time-varying
features (meteorological)*



Model training: uses historical meteorological + geospatial data

Model deployment: uses forecast meteorological + geospatial data

Flood forecasting: scope

- Flash floods — short, intense rainfall events in hilly/urban areas.
- Riverine floods — river overflow from prolonged rainfall or snowmelt.
- Glacial Lake Outburst Floods (GLOFs) — sudden releases from glacier lakes.

Data assembled so far

- Meteorological:
 - ERA5 reanalysis (hourly, downsampled to 6-hour)
 - ECMWF and GraphCastCFS forecasts (see previous slides)
- Hydrological:
 - Observed daily discharge for major Bhutan basins (Lungtenphu, Wangdi Rapid, Pangbang, Tingtibi, Kurjey)
- Flood-specific:
 - Historical flood events (dates, type, region; limited records)
 - 19 known GLOF events (but lots of missing info)
- Geospatial:
 - Elevation, flow accumulation, slope, soil type, land cover (considering HydroSHEDS; may be outdated)
 - Glacier lake characteristics

Spatial alignment challenge

Flood risk modeling faces a core spatial misalignment:

-  Flood dynamics are driven by hydrological units
- (e.g., catchments, river basins, glacier lakes)

-  Meteorological inputs come as gridded data
- (e.g., rainfall, runoff on a regular lat-lon grid)

-  Early warning systems must operate at administrative units
- (e.g., villages, gewogs, dzongkhags — where people live and decisions are made)

Flash flood

Input features:



Static geospatial (catchment-scale)

- DEM-derived slope, elevation, flow accumulation
- Drainage density / distance to stream
- Soil type and infiltration potential
- Land cover (impervious surface, forest, etc.)



Meteorological (time-varying; aggregated to catchment)

- 6-hour cumulative rainfall
- Runoff
- Short-term forecast rainfall (lead 0–24 h)

Modeling Unit: small catchment

- Why: Hydrologically meaningful for flash-flood generation and routing
- Historical flash-flood reports are geocoded and mapped to catchments (from points/admin records → nearest/containing catchment)
- Results can be rolled up to villages/districts for alerts & response

Output Target:

- Flash flood occurrence probability

Flash flood

📦 Example: Feature Table for Small-Catchment Flash Flood Modeling (6-hour resolution)

Catchment_ID	Time	Elev_mean (m)	Slope_std (°)	%Urban	Soil_clay (%)	Rainfall_6h_avg (mm)	Runoff_6h (mm)	Flood_occurred
C001	2025-08-01 00:00	1450	12.3	22.1	30.5	18.4	5.8	Yes
C001	2025-08-01 06:00	1450	12.3	22.1	30.5	12.7	4.1	No
C014	2025-08-01 00:00	1980	5.7	3.8	12.1	4.2	1.3	No
C014	2025-08-01 06:00	1980	5.7	3.8	12.1	9.5	2.6	Yes

- Time: Represents the start time of each 6-hour window
- Rainfall_6h_avg: Mean rainfall over the 6-hour period, aggregated from grid to admin unit
- Static features (elevation, slope, soil, land cover) remain constant across time rows

Riverine flood

Input features:



Static geospatial (basin-scale)

- Basin area and shape
- Elevation and slope (mean or representative)
- Flow accumulation
- Land cover (% forest, urban, agriculture, etc.)
- Soil type or infiltration potential
- Glacier coverage



Meteorological (time-varying; aggregated to river basin)

- Past 3–7 day cumulative rainfall
- Forecasted rainfall
- Runoff
- Short-term forecast rainfall
- Temperature (affecting glacier melt or snowmelt)

Modeling Unit: river basin

- Why: Reflects the natural flow structure of water accumulation and routing
- Results can be rolled up to villages/districts for alerts & response

Output Target:

- River discharge level

Riverine flood

 Example: Feature Table for Riverine Flood Modeling (daily resolution)

Basin	Date	Basin_area (km²)	Elev_mean (m)	Slope_mean (°)	%Forest	Glacier_area (%)	Rainfall_3day_sum (mm)	Runoff_avg (mm/day)	Flood_occurred
X	2025-08-01	320	2850	13.5	48.3	6.4	78.2	12.1	Yes
Y	2025-08-01	145	1680	9.8	21.6	0.0	35.4	5.6	No

- Modeling unit: River basin
- Date resolution: Daily (could also use 6-hour if discharge data supports it)
- Target: Discharge at basin outlet — can be used directly or thresholded to infer flood occurrence

GLOF (Glacial Lake Outburst Flood)

Input Features:



Glacier lake characteristics (static or slowly changing)

- Lake volume and expansion rate (from satellite monitoring)
- Elevation (affects lag time to downstream impact)
- Distance to and width of moraine dam (failure likelihood)
- Seismic activity (possible trigger)
- Downstream exposure (settlements, roads, hydropower, etc.)



Meteorological triggers (time-varying, aggregated to glacier lake level)

- Snowmelt + rainfall combination
- Cumulative rainfall over short period

Modeling Unit: Glacier Lake

- Why: Most predictive features are lake-specific; Risk is tied to individual lake conditions, not entire basins or admin areas
- Results can be rolled up to villages/districts for alerts & response

Output target

- GLOF risk

GLOF (Glacial Lake Outburst Flood)

 Example: Feature Table for GLOF Modeling (6-hour resolution)

Lake ID	Time	Elevation (m)	Volume (MCM)	Moraine_width (m)	Rainfall_6h (mm)	Runoff_6h (mm)	Seismic_event	GLOF_occurred
L001	2025-08-01 00:00	4800	35.2	40	18.4	6.7	No	No
L001	2025-08-01 06:00	4800	35.2	40	28.4	9.1	No	Yes
L002	2025-08-01 00:00	5100	18.7	65	3.5	1.4	No	No
L002	2025-08-01 06:00	5100	18.7	65	10.2	4.9	Yes	Yes

Questions on flood modeling

- We aim to model **three types of floods**. Do you think this is feasible? If not, would combining flash and riverine floods be more practical?
- If time or resources are limited, which flood type should we prioritize for Bhutan?
- Are our current spatial units appropriate for each hazard type? If not, what would you recommend?
- Our current temporal resolution is every 6 hours, with forecasts extending 15–16 days ahead. Any suggestions for improvement or considerations for this setup?
- We have sufficient meteorological data, but are uncertain about the geospatial data requirements—any guidance on what should be included?

Other questions

- Are there any challenges we should anticipate in the next phase of work, particularly related to Bhutan's local geospatial characteristics?
- Are there important datasets for flood modeling that we have not yet identified?
- Of the work we have planned, which components would be most beneficial to local communities if successfully implemented?
- Which output format would be most useful? (Webpage, Mobile app, SMS alerts, Others)